

Dense critical graphs from saturated graphs

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Abstract

Let $f_k(n)$ denote the maximum number of edges in a k -critical graph on n vertices. For every fixed integer $k \geq 8$, we construct a sequence of k -critical graphs G_s with $|V(G_s)| \rightarrow \infty$ and

$$\frac{e(G_s)}{|V(G_s)|^2} \rightarrow \begin{cases} \frac{k-4}{2(k-2)}, & k \text{ even,} \\ \frac{k-5}{2(k-3)}, & k \text{ odd.} \end{cases}$$

For $k = 12$, the density is $2/5 > 3/8$, disproving $f_{12}(n) \sim 3n^2/8$ and hence the general asymptotic formula proposed in Erdős Problem 917. The construction combines a coloring module of Pegden with saturated graphs of sublinear maximum degree.

1 Introduction

All graphs are finite and simple. A graph is k -critical if it has chromatic number k and every proper subgraph is $(k-1)$ -colorable; for graphs without isolated vertices this is equivalent to requiring only that deleting any edge lowers the chromatic number. Write $e(G) = |E(G)|$. For each integer $n \geq 0$, let $f_k(n)$ be the largest edge count of a k -chromatic graph on n vertices satisfying this edge-deletion condition, with $f_k(n) = 0$ if no such graph exists. Erdős [3, p. 28] conjectured that, for fixed $k \geq 6$,

$$f_k(n) \sim \frac{1}{2} \left(1 - \frac{1}{\lfloor k/3 \rfloor} \right) n^2. \quad (1.1)$$

This is the general asymptotic question in Erdős Problem 917 [4]. Dirac [2, p. 90] gave the $k = 6$ construction obtained by joining two odd cycles of equal order, and Erdős [3, p. 28, eq. (3)] observed that it generalizes when $3 \mid k$, giving the coefficient in (1.1). When $3 \nmid k$, Toft's constructions [8] give larger coefficients. For background on dense critical graphs, see Jensen and Toft [6, Section 5.1, pp. 97–98] and Jensen [5]. Luo, Ma, and Yang [7] noted that no constructions improving Toft's asymptotic constants had been found since 1970. We give the following construction.

Theorem 1. *For every integer $k \geq 8$, there is a sequence of k -critical graphs G_s such that $|V(G_s)| \rightarrow \infty$ and*

$$\frac{e(G_s)}{|V(G_s)|^2} \rightarrow \begin{cases} \frac{k-4}{2(k-2)}, & k \text{ even,} \\ \frac{k-5}{2(k-3)}, & k \text{ odd.} \end{cases}$$

For even $k = 2t + 2$, the graph G consists of t copies of Pegden's module $U(S, T)$ [9], with their active sets joined by an almost complete t -partite graph. The omitted edges form a graph Z

whose adjacency is prescribed by a K_t -saturated graph H of small maximum degree from [1]. In a $(2t + 1)$ -coloring, each active set needs three colors (Lemma 2); the degree bound forces two of them to be private to that part. The remaining color must then appear in all t parts, giving a K_t in Z , impossible since H is K_t -free. Deleting an edge between two parts adds an edge to Z , and saturation supplies a K_t through it, allowing one color to be shared in a $(2t + 1)$ -coloring of $G - e$. Complete joins of critical graphs, as in Dirac's construction, cannot share colors between factors; omitting Z costs $o(n^2)$ edges and permits this shared color.

Deleting the $O_k(n)$ module edges leaves a t -partite graph, so the asymptotic density of this construction is at most $(t - 1)/(2t)$, equal to $2/5$ when $k = 12$. For comparison, the upper bound of Luo, Ma, and Yang [7, Theorem 1.1] has coefficient $9/20 - 1/4356 \approx 0.44977$ for f_{12} . Section 4 gives an explicit finite-field realization of the $k = 12$ case.

2 A coloring module

Let $r \geq 3$, let S be r -critical, and let T be $(r - 1)$ -critical. The module $U(S, T)$ consists of disjoint copies of S and T , together with an independent set

$$A = V(S) \times V(T).$$

Each $(x, y) \in A$ is adjacent to $x \in V(S)$ and $y \in V(T)$. There are no edges between S and T , and no other edges. We call A the active set and call $\{x\} \times V(T)$ the row indexed by x .

The following is the $U(r, r - 1)$ case of Pegden's module lemma [9, Lemma 2.5], stated there for triangle-free factors; the lemma holds for general critical factors S and T of the indicated chromatic numbers.

Lemma 2. *Fix a palette of r colors.*

1. *In every proper coloring of $U(S, T)$ from this palette, the active set uses at least three colors.*
2. *Given $a_0 = (x_0, y_0) \in A$ and distinct colors α, β, γ , there is a proper coloring in which the active set uses only α, β, γ , and γ occurs on the active set only at a_0 .*
3. *Given an edge e of $U(S, T)$ and distinct colors α, β , there is a proper coloring of $U(S, T) - e$ in which the active set uses only α, β .*

Proof. For (1), suppose the active colors are contained in $\{\alpha, \beta\}$. Since S uses all r colors, it contains a vertex colored α and a vertex colored β . Every active vertex in the first row must have color β , so no vertex of T can have color β . The second row similarly excludes α from T . This leaves only $r - 2$ colors for T , a contradiction.

For (2), color S so that α occurs only at x_0 . Color T with the $r - 1$ colors other than α , so that β occurs only at y_0 . These colorings exist by criticality: delete the specified vertex, color the remaining graph with one fewer color, and give the deleted vertex its own color. Extend these colorings by

$$\text{col}(x, y) = \begin{cases} \alpha, & x \neq x_0, \\ \beta, & x = x_0, y \neq y_0, \\ \gamma, & (x, y) = (x_0, y_0). \end{cases} \quad (2.1)$$

Each active vertex differs in color from both its neighbors.

For (3), there are four possibilities for the deleted edge. If $e \in E(S)$, color both $S - e$ and T with the $r - 1$ colors other than α , and color all of A with α . If $e \in E(T)$, color $T - e$ with the $r - 2$

colors other than α, β . Choose $x_0 \in V(S)$ and color S with α only at x_0 . Color the row indexed by x_0 with β and all other rows with α .

Suppose instead that e joins an active vertex (x_0, y_0) to one of its two structural neighbors. Use the colorings of S and T from the proof of (2). If the deleted edge joins (x_0, y_0) to x_0 , use (2.1) with the color of (x_0, y_0) changed to α . If it joins (x_0, y_0) to y_0 , change that color to β . In either case the only edge that would become monochromatic is the deleted edge. $\square$

3 From saturated graphs to critical graphs

A graph is K_t -saturated if it contains no K_t , but adding any missing edge creates a K_t . Thus the common neighborhood of every pair of distinct nonadjacent vertices contains a K_{t-2} .

Proposition 3. *Let $t \geq 3$, and let H be a K_t -saturated graph on $v \geq t$ vertices with maximum degree $d < v - 1$. Let $h \geq 2t + 1$ be odd and suppose*

$$v > 2t(t-1)hd. \quad (3.1)$$

Put $a = 2hv$. There is a $(2t+2)$ -critical graph G with

$$|V(G)| = t(a + h + 2v) \quad (3.2)$$

and

$$e(G) \geq \binom{t}{2} a^2 \left(1 - \frac{d}{v}\right). \quad (3.3)$$

Proof. Construction. Set $\ell = 2v$ and take joins of cliques and odd cycles, as in [2, p. 90]:

$$S = K_{2t-2} \vee C_{h-2t+2}, \quad T = K_{2t-3} \vee C_{\ell-2t+3}.$$

Both cycles are odd and have length at least three. Since the join of a p -critical and a q -critical graph is $(p+q)$ -critical, S is $(2t+1)$ -critical and T is $2t$ -critical. (Deleting a vertex or an edge within one factor saves a color there; if a joining edge xy is deleted, give x and y singleton colors in their respective factors and identify those colors.)

Take t disjoint copies $U_1, \dots, U_t$ of $U(S, T)$, with active sets $A_1, \dots, A_t$, each of size $a = h\ell$. In each copy, identify $V(T)$ with $V(H) \times \{0, 1\}$ by any bijection. An active vertex then has the form $(x, (z, \varepsilon))$; give it label z . Let π denote this labeling map. Each label occurs exactly $b = 2h$ times in each active set.

Define a t -partite graph Z on $A_1 \cup \dots \cup A_t$ by putting, for $u \in A_i$ and $w \in A_j$ with $i \neq j$,

$$uw \in E(Z) \iff \pi(u)\pi(w) \in E(H).$$

The graph G retains all the edges of the modules and, between different active sets, takes the complement of Z . There are no other edges between modules. In particular, active vertices with equal labels in different parts are adjacent in G .

We record three properties of Z . First, Z is K_t -free: the labeling map sends every clique injectively to a clique of H . Moreover, adding any missing edge between different parts of Z creates a K_t with one vertex in each part. To see this, let u, w be the endpoints of such a missing edge. If their labels z, z' differ, saturation of H gives a K_{t-2} in $N_H(z) \cap N_H(z')$. Place its labels in the other $t-2$ parts. Together with u, w , they form the required clique in $Z + uw$. If the labels coincide, say both equal z , choose a vertex $z' \neq z$ not adjacent to z ; this is possible because

$d < v - 1$. The same common-neighborhood clique is contained in $N_H(z)$ and gives the required clique.

Second, Z contains a K_{t-1} across any prescribed $t - 1$ parts. Indeed, H has a missing edge; one endpoint together with a K_{t-2} in the common neighborhood gives a K_{t-1} in H . Its labels can be placed in any $t - 1$ parts of Z .

Third, $D := \Delta(Z) = (t - 1)bd = 2h(t - 1)d$, so (3.1) gives $\ell > 2tD$.

The chromatic lower bound. Suppose that G has a $(2t + 1)$ -coloring. Fix a part A_i and any color α . Since its copy of S uses all $2t + 1$ colors, some vertex x of S has color α . The ℓ active vertices in the row indexed by x avoid α , so the pigeonhole principle and $\ell > 2tD$ give a color $\beta \neq \alpha$ occurring more than D times in that row. Applying this observation twice, we may choose two distinct colors α_i, β_i , each used more than D times in A_i .

A color used more than D times in A_i cannot occur in any other active set. A vertex of that color in another part would have to be adjacent in Z to all those more than D vertices, contrary to $\Delta(Z) = D$. Thus the $2t$ chosen colors $\alpha_1, \beta_1, \dots, \alpha_t, \beta_t$ are distinct, leaving a single color γ . The active set A_i can use only $\alpha_i, \beta_i, \gamma$, and Lemma 2(1) forces γ to occur in every part. Choosing a vertex of this color in each part gives a K_t in Z , a contradiction. Thus $\chi(G) \geq 2t + 2$.

Colorings after edge deletion. Use the $(2t + 1)$ -color palette

$$\alpha_1, \beta_1, \dots, \alpha_t, \beta_t, \gamma.$$

The pair α_i, β_i will be used only in A_i among the active sets. Structural vertices may use the full palette, since they have no neighbors in other modules.

Let $e = uw$ join two different active sets in G . By the first property of Z , there is a K_t in $Z + uw$ containing uw , with vertices $a_i \in A_i$. Apply Lemma 2(2) in each module, using active colors $\alpha_i, \beta_i, \gamma$ and assigning γ only to a_i in A_i . The only color shared by different active sets is γ . Its t vertices are pairwise nonadjacent in $G - e$, because they form a clique in $Z + uw$. We have obtained a $(2t + 1)$ -coloring of $G - e$.

Now let e lie within U_i . Apply Lemma 2(3) to color $U_i - e$, using only α_i, β_i on A_i . Choose a K_{t-1} in Z across the other $t - 1$ parts. In each of those modules, apply Lemma 2(2) with the selected vertex as the only active vertex of color γ . Once again this is a proper $(2t + 1)$ -coloring of $G - e$.

These cases cover every edge of G . Recoloring one endpoint of a deleted edge with a new color gives $\chi(G) \leq 2t + 2$. Since G has no isolated vertices, every proper subgraph of G lies in some $G - e$. Hence G is $(2t + 2)$ -critical.

Counting vertices and edges. Each module has $a + h + 2v$ vertices, giving (3.2). Between a fixed pair of active sets, Z has $2b^2m$ edges: each edge of H gives two ordered pairs of labels, with b^2 pairs of active vertices for each, where $m = e(H)$. Since $2b^2m \leq b^2vd$ and $a = bv$, at least $a^2(1 - d/v)$ edges remain between each pair of active sets, proving (3.3). $\square$

Remark (Exact edge count). The construction in Proposition 3 satisfies

$$e(G) = \binom{t}{2}(a^2 - 8h^2m) + t(2a + (2t - 1)h + 4(t - 1)v - (4t^2 - 4t - 1)), \quad (3.4)$$

where $m = e(H)$. Indeed,

$$e(S) = (2t - 1)h - (t - 1)(2t + 1), \quad e(T) = 4(t - 1)v - t(2t - 3),$$

and each module has $2a$ edges incident with its active set.

Proof of Theorem 1. First let $k = 2t + 2$ be even, where $t \geq 3$ is fixed. By [1, Theorem 7], there are K_t -saturated graphs H of order $v \rightarrow \infty$ and maximum degree $d = O_t(\sqrt{v})$. Choose

$$h = 2\lfloor v^{1/4} \rfloor + 1.$$

Then $h \rightarrow \infty$, $h = o(\sqrt{v})$, and $2t(t-1)hd/v \rightarrow 0$. Consequently $h \geq 2t + 1$, $v \geq t$, $d < v - 1$, and (3.1) hold for all sufficiently large v .

Apply Proposition 3, and write $a = 2hv$ and $N = |V(G)|$. We have

$$\frac{N}{ta} = 1 + \frac{1}{h} + \frac{1}{2v} \rightarrow 1, \quad \frac{d}{v} \rightarrow 0.$$

The edges within modules number $O_t(a)$, by (3.4), and hence

$$\binom{t}{2} a^2 \left(1 - \frac{d}{v}\right) \leq e(G) \leq \binom{t}{2} a^2 + O_t(a).$$

It follows that

$$\frac{e(G)}{N^2} \rightarrow \frac{\binom{t}{2}}{t^2} = \frac{t-1}{2t} = \frac{k-4}{2(k-2)}.$$

For odd $k = 2t + 3 \geq 9$, join each graph in the even construction to K_1 . The resulting graph is k -critical, has $N + 1$ vertices and $e(G) + N$ edges, and has the same limiting density, now equal to $(k-5)/(2(k-3))$. $\square$

4 An explicit family for $k = 12$

We use the $k = 5$ case of [1, Section 4, Example 2] and include the saturation verification in Lemma 4.

Lemma 4 (Alon–Erdős–Holzman–Krivelevich). *For every prime power $Q \geq 4$, there is a K_5 -saturated graph H_Q with*

$$|V(H_Q)| = 13Q^2 + 12Q, \quad \Delta(H_Q) = 22Q - 3. \quad (4.1)$$

Proof. We give the verification of Example 2 in [1] for $k = 5$. Let F be a field of order Q . The levels are $F \cup \{\infty\}$, given any linear order, and the places in each level are indexed by $\mathbb{Z}/Q\mathbb{Z}$ through a fixed bijection with F . For each $(a, b) \in F^2$, form a line $L_{a,b}$ whose point in finite level u has field coordinate $au + b$, and whose point in level ∞ has coordinate a . Every line has one point in each level, every point lies on Q lines, and any two distinct lines meet: if their slopes agree they meet in level ∞ ; otherwise solve $(a - a')u = b' - b$.

A core vertex is a quadruple (i, j, t, c) , where i is its level, $j \in \mathbb{Z}/Q\mathbb{Z}$ its place, $t \in \mathbb{Z}/3\mathbb{Z}$ its type, and $c \in \{0, 1, 2, 3\}$ its copy. In one level, two core vertices are adjacent precisely when their copies differ and their (j, t) pairs differ. Between levels $i < i'$, adjacency means

$$t' = t + 1, \quad j' \in \{j + 1, j + 2, j + 3\}.$$

Finally, add one vertex for each line, adjacent to all twelve core vertices at each of its points. The line vertices are independent. All additions in the place coordinate are modulo Q , irrespective of the characteristic of F .

Excluding K_5 . A triangle of core vertices cannot occupy three levels: its edges would require both $t'' = t + 1$ and $t'' = t + 2$ modulo three. Thus every core clique occupies at most two

levels. Within one level it has at most four vertices, since copies must differ. If it meets two levels, all vertices in each level have the same type, so their places and copies are distinct. The three successors of a place form a proper cyclic interval, since $Q \geq 4$, and different places give different successor triples. The same is true of predecessor triples. Hence two distinct places have at most two common successors or predecessors. A clique meeting two levels therefore cannot have three vertices in one level and two in the other, nor four in either level.

The neighborhood of a line vertex is K_4 -free. Indeed, a clique in that neighborhood occupying one level has at most three vertices, as its places agree and its types must differ. If it occupies two levels, the preceding same-type observation permits at most one vertex in each. Since line vertices are independent, this also excludes every K_5 containing a line vertex.

Saturating every missing edge. It suffices to exhibit a triangle in the common neighborhood of any two distinct nonadjacent vertices x, y . If both are line vertices, take their intersection point and three core vertices there with distinct types and copies. If x is a core vertex and y a nonincident line, take the point of y in the level of x . Its place differs from that of x ; use all three types there, in the three copies other than the copy of x .

Suppose next that both vertices are in the same level. If their copies agree, choose a place different from both of their places, and use its three types in the three other copies. Such a place exists because $Q \geq 4$. If their copies differ, nonadjacency means that their places and types agree. Choose a line through that point and two core vertices there using the other two types and the other two copies. These three vertices give the triangle.

Finally, suppose x lies below $y = (i', j', t', c')$. In the level of x , take the three places $j' - 1, j' - 2, j' - 3$, all of type $t' - 1$, assigning them the three copies different from the copy of x . They form a triangle and are adjacent to y . None has the same place and type as x , since that would make x adjacent to y . Thus they are also adjacent to x . This exhausts the missing edges.

Order and maximum degree. There are $12Q(Q + 1)$ core vertices and Q^2 line vertices. A core vertex has $3(3Q - 1)$ neighbors in its level, $12Q$ in the other levels, and Q line neighbors. Its degree is therefore $22Q - 3$. A line vertex has degree $12(Q + 1) < 22Q - 3$ for $Q \geq 4$, proving (4.1). In particular,

$$\frac{\Delta(H_Q)}{|V(H_Q)|} \leq \frac{22}{13Q} \rightarrow 0 \quad (Q \rightarrow \infty).$$

□

The case $k = 12$. For each integer $s \geq 4$, set

$$h = 3^s, \quad Q = h^2,$$

and take $H = H_Q$ from Lemma 4. Write $v = |V(H)|$ and $d = \Delta(H)$. Then

$$v = 13h^4 + 12h^2, \quad d = 22h^2 - 3 < v - 1.$$

The integer h is odd and at least 81. Moreover,

$$40hd < 880h^3 < 13h^4 \leq v,$$

since $13h \geq 13 \cdot 81 > 880$. Proposition 3 with $t = 5$ gives a twelve-critical graph G_s of order $N_s = 5(2hv + h + 2v)$. Since $N_s/(10hv) \rightarrow 1$ and $d/v \rightarrow 0$, (3.3) and (3.4) give $e(G_s)/N_s^2 \rightarrow 2/5$. Finally, $N_s \rightarrow \infty$ and $f_{12}(N_s) \geq e(G_s)$, so

$$\limsup_{n \rightarrow \infty} \frac{f_{12}(n)}{n^2} \geq \frac{2}{5} > \frac{3}{8}.$$

Thus $f_{12}(n) \not\sim 3n^2/8$.

□

Code availability

A Lean 4 formalization of the $k = 12$ case of Theorem 1, Lemma 4, and the corresponding instances of Lemma 2 and Proposition 3 is available at <https://github.com/FireflySentinel/erdos-917>. The general result for other chromatic numbers is not formalized.

Declaration of generative AI and AI-assisted technologies in the writing process

GPT-6 Astra proposed the construction of critical graphs and the density estimates, drafted the manuscript, and generated the Lean formalization. GPT-5.6 Sol and Claude Opus 5 were used to organize earlier research material and to review the Lean code. The author checked the arguments, completed the final manuscript, and takes full responsibility for the content.

References

- [1] N. Alon, P. Erdős, R. Holzman, and M. Krivelevich, On k -saturated graphs with restrictions on the degrees, *J. Graph Theory* **23** (1996), 1–20. <https://web.math.princeton.edu/~nalon/PDFS/aehk1.pdf>.
- [2] G. A. Dirac, A property of 4-chromatic graphs and some remarks on critical graphs, *J. London Math. Soc.* **27** (1952), 85–92. doi:10.1112/jlms/s1-27.1.85.
- [3] P. Erdős, Problems and results in chromatic graph theory, in *Proof Techniques in Graph Theory* (Proc. Second Ann Arbor Graph Theory Conf., Ann Arbor, 1968), Academic Press, New York, 1969, 27–35. https://www.renyi.hu/~p_erdos/1969-13.pdf.
- [4] T. F. Bloom, *Erdős Problem #917*, <https://www.erdosproblems.com/917>. Accessed 5 September 2026.
- [5] T. R. Jensen, Dense critical and vertex-critical graphs, *Discrete Math.* **258** (2002), 63–84. doi:10.1016/S0012-365X(02)00262-5.
- [6] T. R. Jensen and B. Toft, *Graph Coloring Problems*, Wiley-Interscience, New York, 1995. Chapter 5: Critical graphs.
- [7] C. Luo, J. Ma, and T. Yang, On the maximum number of edges in k -critical graphs, *Combin. Probab. Comput.* **32** (2023), 900–911. doi:10.1017/S0963548323000238.
- [8] B. Toft, On the maximal number of edges of critical k -chromatic graphs, *Studia Sci. Math. Hungar.* **5** (1970), 461–470.
- [9] W. Pegden, Critical graphs without triangles: an optimum density construction, *Combinatorica* **33** (2013), 495–512. arXiv:1101.4417v2.